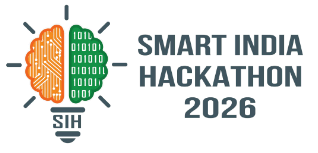


SMART INDIA HACKATHON 2026

IDEA SUBMISSION



- **Problem Statement Title:** **Last-Mile Delivery Verification and Audit Platform for Disaster Early Warnings**

Ministry of Home Affairs — National Disaster Management Authority (NDMA) · Theme: Disaster Management

- **PS Category- Software/Hardware:** **Software**

- **Team Name:** **Fourth Principle**

Jaypee Institute of Information Technology

- **Team Members Names:**

1. Sasmit Yadav (Team Leader) 2. Meghna Tomar 3. Yug Thakral
4. Shreya Goel 5. Gautam Garg 6. Harsh Vardhan Singh



42

features specified

6,836

sub-districts targetable

22

scheduled languages

6

tiers of delivery evidence

0

govt integrations to start

SETU

India already tells a billion people that a disaster is coming.
Nothing tells you whether one of them heard it. **That is the layer we built.**

◆ Proposed Solution (Describe your Idea/Solution/Prototype)

- Detailed explanation of the proposed solution
 - SETU is the delivery, acknowledgement, response and audit layer beneath SACHET and IMD. It does not issue warnings — it proves whether one arrived.
 - INGEST — USGS + GDACS, zero-auth, verified live · plus our own thunderstorm nowcast, because no free feed covers convective weather.
 - TARGET — PostGIS over 6,836 sub-districts; population and building exposure previewed before dispatch.
 - AUTHORISE — 6 automated checks, then 2 humans for an extreme alert. A USGS quake needs none: the seismograph is the second pair of eyes.
 - DELIVER — push · SMS · IVR · siren · peer relay. Retry and escalation policy-driven; not one timing hardcoded.
 - PREDICT — a reach-failure model escalates *before* the towers fall, not after a retry times out.
 - ASSURE — 6 evidence tiers. Where a channel cannot prove a tier, it says so, and says why.
 - RESPOND — “I’m safe” or Trapped / Medical / Cannot evacuate — one tap, or one keypress on a feature phone.
 - RELAY — every channel exhausted → the system calls a person: panchayat, police, ASHA.
 - PROVE — append-only hash-chained ledger; immutability enforced by a Postgres trigger, not by our code.
- How it addresses the problem
 - Wayanad, 30 July 2024 — 400+ dead, 118 never found. The Union Home Minister told Parliament **four warnings** went to Kerala between 23 and 26 July. The Chief Minister replied the red alert came *after* the landslides. **Both statements are on the record. Nothing in the chain can settle which is true.**
 - Cyclone Ockhi, Nov–Dec 2017 — 318 dead, 141 missing. IMD had issued a *fishing advisory*, not a cyclone warning. The state blamed the centre; the centre’s agency blamed the storm’s stage. **The boats were already at sea — no channel existed that could confirm anyone aboard heard anything.**
 - Irshalwadi, Raigad, 19 July 2023 — 26 dead, 109 missing. An orange alert stood over the district for two days. The hamlet sits **2.4 km beyond the nearest road**. The alert was issued. Whether it travelled that last 2.4 km is unrecorded.
 - Three events, three decades of forecasting improvement, **one identical hole**: India’s forecasting is not the failure point — **the last fifty kilometres is**, the one link with no instrumentation at all.
 - SETU makes that link measurable: per recipient, per channel, per tier, on the record.

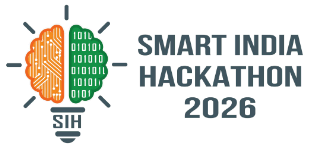
• Innovation and uniqueness of the solution

CAPABILITY	SACHET / IMD today	SETU
Publishes and geo-targets a warning	✓	✓
Per-recipient delivery record	✗	✓
Retry and automatic channel escalation	✗	✓
Predicts reach failure <i>before</i> sending	✗	✓
Two-person authorisation on an extreme alert	✗	✓
Evidence tiers — and states what it cannot prove	✗	✓
Two-way: citizen can answer “I need help”	✗	✓
Human relay when every channel is exhausted	✗	✓
Tamper-evident audit trail	✗	✓

Checked against SACHET’s public citizen-facing surface only — stated as what we found no evidence of.



TECHNICAL APPROACH



• Technologies to be used (e.g. programming languages, frameworks, hardware)

FRONTEND

React · TypeScript · Vite · Tailwind · MapLibre GL + pmtiles · TanStack Query & Virtual · Workbox PWA · Web Bluetooth

BACKEND

Python · FastAPI · Pydantic v2 · APScheduler · SQLAlchemy async / asyncpg · Alembic · slowapi

DATA & GEOSPATIAL

PostgreSQL + PostGIS + pgvector (Neon) · Redis Streams (Upstash) · geoBoundaries ADM3/ADM5 · WorldPop · Google Open Buildings · OSM Overpass · Copernicus GLO-30 DEM · OpenCellID · Protomaps basemap

AI & MACHINE LEARNING

sentence-transformers MiniLM-L12-v2 · LightGBM · AI4Bharat IndicTrans2 dist-200M (22 languages) · pgvector RAG

CHANNELS & CRYPTOGRAPHY

Firebase FCM · Twilio SMS & IVR · Brevo email · webhook siren · HumanRelayAdapter · Ed25519 signing (PyNaCl / @noble)

DEVOPS & TESTING

Docker Compose · GitHub Actions · Vercel · Render · Hugging Face Spaces · Hypothesis · Playwright · Locust · Sentry



- Every limit verified by **directly fetching the source, twice** — not read off a marketing page.
- Pilot runs on free tiers with no procurement. **We do not claim free scales forever** — national volume is a paid-tier migration, not a rewrite.
- **The basemap is a file we host, not a service we call** — the map still draws with the cable pulled.

95%

branch coverage gate
on the state machine

6,836

units targetable
nationwide

42

features specified
28 in the build window

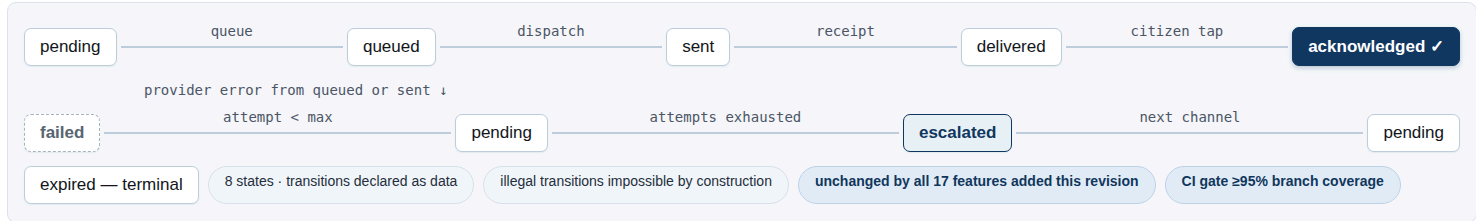
0

new models added
by the new layer

• Methodology and process for implementation (Flow Charts/Images/working prototype)

- 1 **INGEST** — adapters load from a database table, never imported by name. A new feed is one INSERT.
- 2 **DE-DUPLICATE & VERSION** — clustered only if areas *and* time windows overlap; attached to one incident as v1 → v2 → v3.
- 3 **TARGET & SCORE** — polygon intersected against admin geometry; exposure previewed; each unit scored for reach failure.
- 4 **VALIDATE** — 6 rules: geometry, expiry, target count, escalation policy, translation. A failure **blocks dispatch and names itself**.
- 5 **AUTHORISE** — 2 distinct officers, enforced by a unique index. Machine-origin alerts record source provenance instead.
- 6 **DISPATCH** — batched fan-out keeps a 340-unit alert at four queue writes; channel by severity *and* predicted risk.
- 7 **ASSURE** — 6 tiers from real signals: FCM id, service-worker callback, carrier receipt, answered call. Unprovable tier → **“not applicable”**.
- 8 **RESPOND & RELAY** — safe / trapped / medical / cannot-evacuate → weighted queue. Channels exhausted → **call a person**.
- 9 **PROVE** — every state change writes its audit row *in the same transaction*. The ledger cannot drift.

Per-recipient delivery state machine



Machine learning — five models, each with its own evidence standard

MODEL	TECHNIQUE	WHAT WE REPORT
De-duplication CORE	Multilingual embeddings → clustering, vetoed unless areas intersect and windows overlap	Precision / recall / F1 on ~200 hand-labelled real alert pairs, held-out split
Reach-failure CORE	LightGBM on terrain, tower density, rainfall, severity, hour — every feature measured	Badged bootstrap ; validated on Wayanad and Irshalwadi, n = 2 . The new tiers build the dataset that fixes it
Translation CORE	IndicTrans2 dist-200M — distilled, because the 1B card recommends a GPU we lack	Cached per alert and language; serves app, IVR script and relay call from one cache
Nowcast CORE	CAPE, Lifted Index, CIN, precip probability → risk at CAPE > 1000 J/kg , LI < -2	Meteorological standard, not tuned to pass a demo. Non-authoritative — cannot authorise its own alerts
Copilot PHASE 2	RAG over NDMA SOPs and dam Emergency Action Plans in pgvector; citations mandatory	Below a confidence floor: <i>“I don’t have a document covering that”</i>

The 17 features added in this revision introduced **zero new models** — every new decision is a published formula whose inputs are stored beside its output.

• Analysis of the feasibility of the idea

TECHNICAL Polygon intersection, a work queue, a state machine, a hash chain — all solved primitives. **No new research required.** Every dependency verified by direct fetch, twice; the newest layer added **no new dependency at all.**

ECONOMIC ₹0 for the pilot — and no procurement to start it. We do not claim free scales forever: national volume is a paid-tier migration on the same architecture, not a rewrite.

OPERATIONAL Additive, not disruptive — consumes existing feeds, instruments the layer *below* them. A new source or channel is one **INSERT**: no code change, no redeploy.

LEGAL & ETHICAL ODbL / CC-BY / MIT only — GADM rejected on its licence. Consent enforced by a database constraint. A relay volunteer receives **a count and an area, never a list of who asked for help.**

DELIVERY Six owners · 28 core features · five go/no-go gates. Feature freeze enforced by a repository workflow, **not by memory.**

SIZED AGAINST REAL LIMITS, NOT ASSUMED ONES

- Batched fan-out: ~**150–310** queue commands per alert run instead of ~**5,000** — about **fifty full runs a day** inside a 500,000/month quota.
- Two transformer models cannot fit beside the API in 512 MB → ML service moves to a larger free CPU host. **Found by sizing on day one.**
- Village geometry nationwide is **1.01 GB** against a **0.5 GB** free database → sub-district nationwide + village level in two states. **A hosting cost, not a technical barrier.**
- **A hosted basemap would go blank when we unplug the network** → the map is a file we ship, not a service we call.

EXECUTION PLAN — FIVE GO/NO-GO GATES, FREEZE ENFORCED BY THE REPOSITORY

- | | | | | | |
|---|--|---|--|---|--|
| 1
Day 1
feasibility spike
no product code | 2
Day 3
one real alert row,
real adapter | 3
Day 6
end to end;
cable test begins | 4
Day 9
21-step run,
one unbroken take | 5
Day 11
feature freeze,
absolute |
Day 13
six rehearsals,
all six can drive |
|---|--|---|--|---|--|

• Potential challenges and risks

Nationwide SMS is legally unavailable to us — TRAI DLT sender registration needs a registered legal entity and ~10 working days.

IMD's public API returns 401 on every endpoint, and SACHET publishes no documented alert-discovery feed.

The reach-failure model has no history to learn from — no system has ever recorded acknowledgement outcomes.

Peer relay could be used to inject a fake evacuation order — it reaches a phone without touching our server.

An authorisation gate could delay a life-safety alert — the safety feature becoming the hazard.

Free-tier quota, a cold start, or an exhausted trial credit bites mid-evaluation.

The network fails during judging — or scope creep leaves an unrehearsed demo.

All 42 features are specified, owned and scheduled. 28 land inside the thirteen-day build window; the remaining 14 are sequenced immediately behind them, each independently deliverable. Every threshold, metric and limitation is published by the system itself at `/api/v1/methodology`.

• Strategies for overcoming these challenges

→ Real SMS to verified numbers in the room. Every other recipient runs the *identical* engine against a simulated carrier, flagged `simulated=true` and badged **SIM** on screen. **We never pretend.**

→ Both demoted to optional upgrades. USGS, GDACS and our own nowcast already satisfy ingestion; either slots in later with **one INSERT**. Nothing load-bearing waits on an approval we do not control.

→ Trained on a *published* physical failure process using real terrain, tower-density and rainfall features; badged **bootstrap**. Validated on Wayanad and Irshalwadi — **we say n = 2 out loud.**

→ Every alert **Ed25519-signed server-side**, verified on the device **before anything renders**; expiry sits inside the signed payload. An unverified payload is discarded silently — a "suspicious alert" banner is its own panic vector.

→ It applies where a *human* composed the alert. A USGS earthquake dispatches with zero human steps. And we measure our own gate: `approval_wait_seconds`, p50 and p95, on our dashboard.

→ Batched writes · daily command gauge alarming at 80% of budget · keep-alive against idle suspension · local Redis for all development · voice beats rehearsed on the simulated adapter.

→ The offline case *is* the demo: frozen nightly fixtures — including every new table — and the cable pulled every day from day six. Build order fixed in writing, closed by a repository-enforced freeze.

IMPACT AND BENEFITS



• Potential impact on the target audience

DISTRICT EMERGENCY OFFICER

Sees *during* the event which units are silent — and **who inside them asked for help**, ranked and explained.

STATE DMA / NDMA

Reach by district and channel, replacing a dispatch count that never measured arrival. Plus a standing map of structurally unreachable villages.

CITIZEN IN THE HAZARD ZONE

A warning in their own language that survives the network going down.
And **a way to answer that they are not safe** — by tap or keypad.

FIELD RESPONDER / RELAY NODE

Told **how many households were unreachable, and where**.
Confirms with one keypress. Never given a list of who asked for help.

AUDITOR / RTI APPLICANT

"Did the warning arrive?" becomes a database query, not a two-year inquiry.
A reconstructable timeline, **without the personal data the record protects**.

TODAY

- Broadcast to a region — no per-person record
- Nobody had to authorise it
- A trapped citizen has no way to say so
- After the event, the question has no data behind it
- Accountability ends in dispute; the next district learns nothing

WITH SETU

- Per-recipient trail: attempted → accepted → delivered → opened → acknowledged → answered
- Hash-chained and tamper-evident
- The authorisation that released it, attached
- The help that was requested, attached
- **The answer exists before the question is asked**

6,836

sub-districts
targetable nationwide

22

scheduled languages
at delivery

0

government integrations
needed to start

100%

of state changes
carry an audit row

FROM THIS HACKATHON TO NATIONAL SCALE — NO NEW LAW, NO NEW BUDGET

1 PILOT

Two districts, live global feeds. **Zero government integration required** — runs before anyone grants us anything.

2 STATE

A State DMA registers its CAP feed, officer roles and relay nodes — one database row each. Quarantine view shows whether its feed is even well-formed.

3 NATIONAL

SACHET and IMD become two more sources. SETU becomes the accountability layer under a system already reaching a billion people.

The duty exists. The feeds exist. The handsets exist.
What is missing is the record — and the answer back.

• Benefits of the solution (social, economic, environmental, etc.)

SOCIAL

- Instruments the last fifty kilometres — where Wayanad, Ockhi and Irshalwadi actually failed.
- **Makes the warning two-way** — one tap, or one keypress on a ₹1,000 handset, becomes a prioritised case with a named responder.
- 22 scheduled languages, cached — one translation serves the app, the voice call and the relay briefing.
- Works when the towers do not — held on the handset, acknowledgement queued, and a neighbour's phone can carry it the last hundred metres.
- Consent explicit; a push token is enough to register. The unregistered are reached by area broadcast and by a person — never by misusing data we were not given.

ECONOMIC

- No procurement cycle, no licence, no new budget line to run the pilot.
- At national scale: a paid-tier migration on the same architecture, not a rewrite.
- Costly channels — SMS, IVR, a human's time — spent **only where reach is predicted to fail**.
- Reuses SACHET and IMD rather than replacing them; no existing public investment is discarded.
- One deployment serves every state authority; each new one onboards as a database row, not a tender.

ENVIRONMENTAL

- Polygon-accurate targeting replaces blanket broadcast — less alert fatigue, fewer wasted messages.
- Digital acknowledgement removes the paper trail that post-event verification otherwise generates.
- Adds a convective-hazard class no free feed currently serves for India, from open atmospheric data alone.
- Runs entirely on shared free-tier infrastructure — not one new server is commissioned for the pilot.

TECHNOLOGICAL & GOVERNANCE

- **Creates the acknowledgement dataset India does not have** — every future model of warning reach needs it.
- **No high-severity alert leaves without a named authorisation**. Our own nowcast model does not get to authorise itself.
- Publishes every threshold, metric and limitation as a live endpoint — **and every signal it refuses to claim, with the reason**.
- **Names the villages where the last resort does not exist yet** — turning a post-mortem finding into a preparedness task.
- Turns a life-safety duty into a tamper-evident, RTI-answerable record enforced by the database itself.

Wayanad, Ockhi, Irshalwadi — 740+ dead, 360+ never found, and in every one of them whether the warning arrived is still disputed. SETU does not promise a better forecast. It promises that the next dispute has an answer.

• Details / Links of the reference and research work

1 NDMA SACHET — Common Alerting Protocol platform & Integration Guide for Agencies

[sachet.ndma.gov.in](https://sachet.ndma.gov.in/docs/Integration_Guide_For_Agencies.pdf) · [/docs/Integration_Guide_For_Agencies.pdf](#)

The system SETU sits beneath. Documents FetchXMLFile with ETag polling — and no per-recipient acknowledgement surface we could find on its public side.

3 USGS Earthquake Catalog — FDSN Event Web Service

earthquake.usgs.gov/fdsnws/event/1/

Primary ingestion source. Live India-region seismic events, zero authentication, confirmed by direct fetch. Flagged *authoritative*, so its alerts dispatch without a human wait.

5 Open-Meteo — CAPE, Lifted Index, Convective Inhibition, precipitation probability

open-meteo.com/en/docs

Free and keyless, global including India, hourly and 15-minute. The input to the thunderstorm nowcast classifier we build ourselves.

7 geoBoundaries gbOpen — India ADM3 / ADM4 / ADM5 (ODbL 1.0)

geoboundaries.org

6,836 sub-districts at 38 MB; 649,771 villages at 1.01 GB — the measurement behind our resolution decision. **GADM was rejected**: its licence forbids redistribution.

9 AI4Bharat IndicTrans2 — en-indic-dist-200M (MIT)

huggingface.co/ai4bharat/indictrans2-en-indic-dist-200M

Translation across 22 scheduled languages. The distilled variant was chosen deliberately: the 1B model card recommends a GPU, and our free host has none.

11 Copernicus GLO-30 DEM (AWS Open Data) · WorldPop · Google Open Buildings · OpenCelliD

registry.opendata.aws/copernicus-dem · hub.worldpop.org · opencellid.org

Terrain ruggedness, population, building count and tower density — every input feature of the reach-failure model is measured, not assumed.

13 Protomaps / PMTiles — OpenStreetMap basemap, self-hosted (ODbL 1.0)

protomaps.com · openstreetmap.org/copyright

A single tile file, not a hosted service. **A hosted basemap goes blank the moment the network is unplugged** — and the offline map is the demo.

2 India Meteorological Department — public API reference

api.imd.gov.in/public/api_reference.html

Twenty documented endpoints. `districtwarning` and `cyclone_track` both returned **HTTP 401** when called directly — the reason IMD is an upgrade track, not a dependency.

4 GDACS — Global Disaster Alert and Coordination System (EC JRC / UN OCHA)

[gdacs.org](https://gdacs.org/gdacsapi/api/events/geteventlist) · [/gdacsapi/api/events/geteventlist](#)

Primary ingestion source for cyclone, flood, wildfire and drought. Live JSON, zero authentication, confirmed by direct fetch.

6 OASIS Common Alerting Protocol v1.2 + Disaster Management Act, 2005 & NDMA Guidelines

docs.oasis-open.org/emergency/cap/v1.2 · ndma.gov.in

The interchange standard 36 state authorities implement independently — hence a deliberately defensive parser. And the statutory duty to warn that SETU makes measurable.

8 OpenStreetMap Overpass API — shelters, schools, community centres, hospitals

wiki.openstreetmap.org/wiki/Overpass_API

The location data behind “nearest safe zone”. Free, no authentication, with a published fair-use policy our query volume sits far inside.

10 TRAI DLT registration — via Twilio's India messaging guidelines

twilio.com/en-us/guidelines/in/sms

The documented source of our SMS constraint: sender-ID registration on the operators' DLT portal requires a registered legal entity and roughly ten working days.

12 Wayanad, 30 July 2024 · Cyclone Ockhi, Nov–Dec 2017 · Irshalwadi (Raigad), 19 July 2023

[the three documented cases behind this submission](#) · [Wayanad + Irshalwadi validate the reach-failure model \(n = 2\)](#)

Wayanad: the Home Minister cited four warnings sent 23–26 July; the Chief Minister said the red alert followed the landslides. Ockhi: a fishing advisory, not a warning, to boats already at sea. Irshalwadi: an orange alert for two days over a hamlet 2.4 km past the nearest road. **Each is a warning issued, a delivery disputed, and no record able to settle it.**

14 Ed25519 signatures (RFC 8032) · PyNaCl · Web Bluetooth GATT

datatracker.ietf.org/doc/html/rfc8032 · developer.mozilla.org/Web_Bluetooth_API

Peer relay reaches a phone without touching our server, so every alert is signed and verified **before it renders**. Two limits we state rather than discover: Chromium-only, and no background scanning — so relay is one hop, on a deliberate tap.

Every source above was verified by directly fetching or querying it, in two passes.

An unconfirmed source is marked unverified and carries a named fallback — never a silent assumption.

Where a signal does not exist at all — an SMS read receipt, a siren's delivery confirmation — we publish the reason instead of a number.